VISIONINSPECT AI

Milestone 3 – Defect Classification, Severity Assessment, Quality Control and Analytics

Milestone	Milestone 3
Focus	Extend the inspection engine into a complete quality-control platform with classification, severity, analytics and reporting.

1. Introduction

Milestone 3 focused on expanding the AI inspection workflow into a complete manufacturing quality-management platform. The work included defect classification, severity assessment, quality decisions, inspection history, analytics dashboards and report generation.

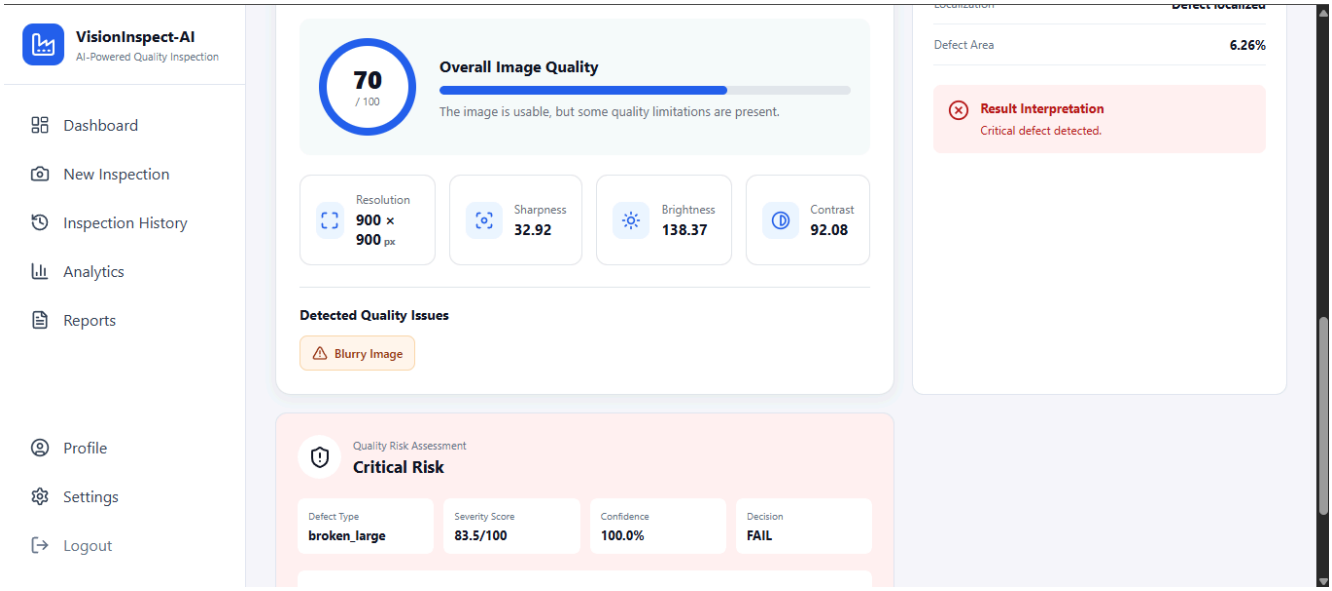
2. Milestone 3 Objectives

- Implement defect classification for inspected products.
- Develop severity assessment for detected defects.
- Generate Pass, Fail and Review quality decisions.
- Persist and display inspection history.
- Develop manufacturing analytics dashboards.
- Implement defect and severity distributions and inspection trends.
- Provide production-quality reports and CSV export.
- Support role-oriented Quality Engineer and Factory Supervisor workflows.

3. Defect Classification

After anomaly detection, the application uses category-specific reference information to identify the most likely defect type. Classification results are included in the inspection record and displayed to the user.

- Category-specific normal reference images are used by the classification workflow.
- Prototype/reference information is used to estimate the defect class.
- The predicted defect type is stored with the inspection record.
- Classification results are visible in Inspection History and analytics.

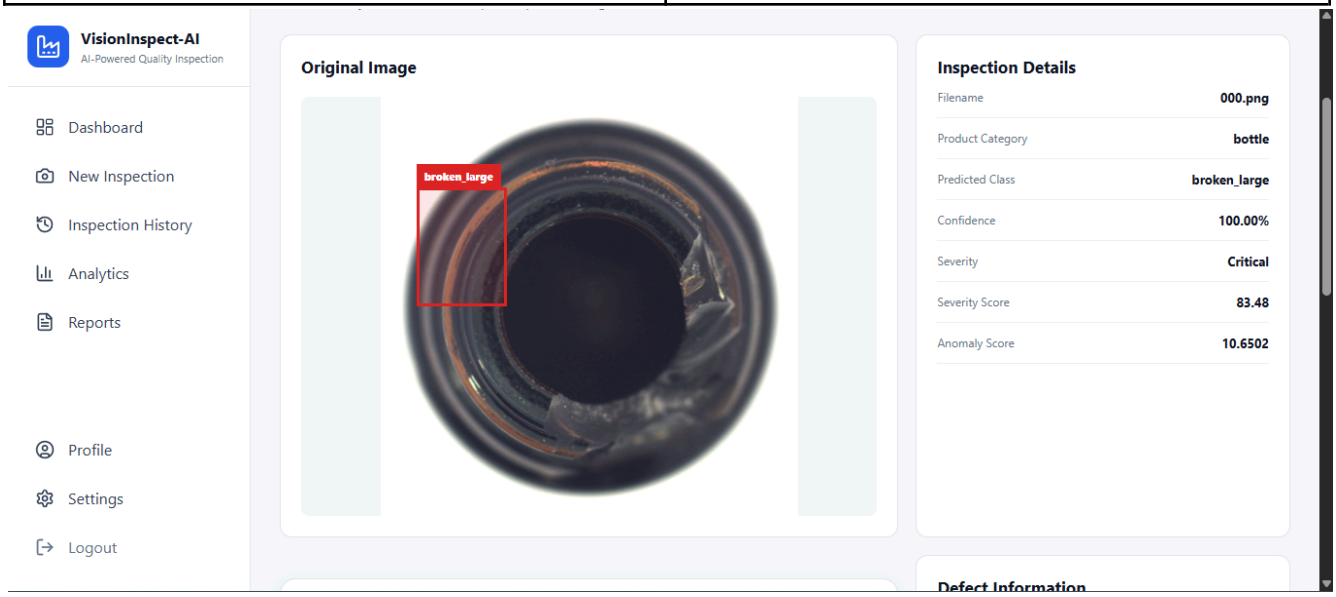


4. Severity Assessment

The project implements a severity framework that combines multiple factors to support consistent quality assessment. The defined weighting is Size 30%, Location 25%, Defect Type 25% and Confidence 20%.

Factor	Weight
Defect Size	30%
Defect Location	25%
Defect Type	25%
Model Confidence	20%

Severity Level	Score Range
Critical	80–100
High	60–79
Medium	40–59
Low	0–39



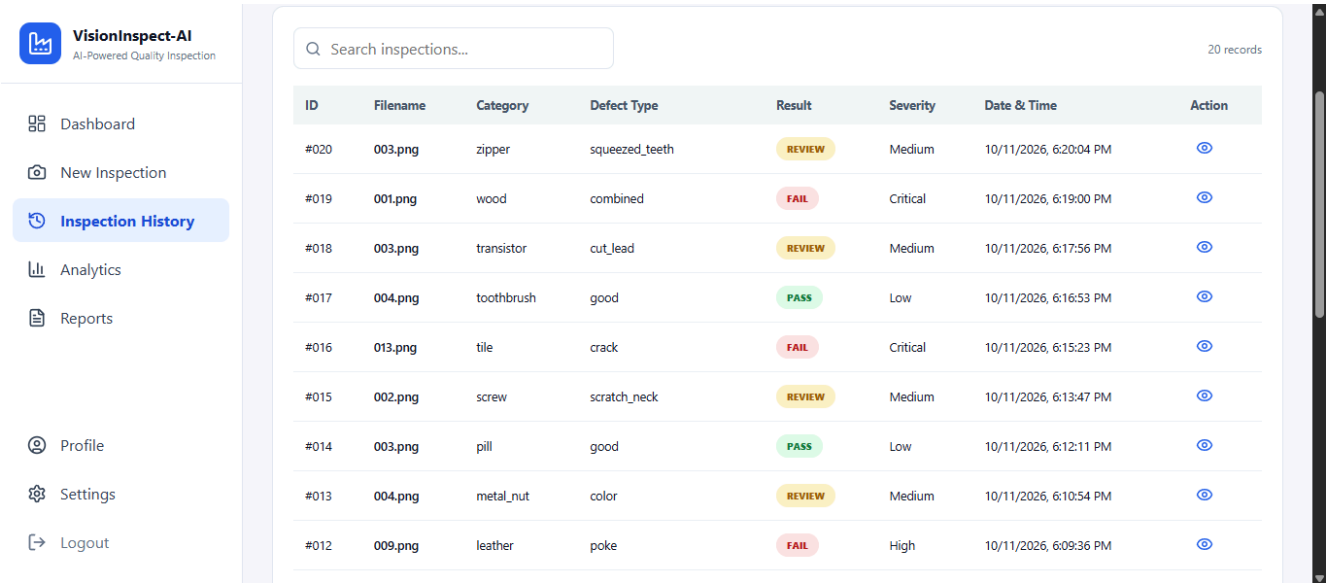
5. Quality Control Decision

The inspection result is translated into a quality decision that can be displayed as PASS, FAIL or REVIEW. This enables the system to support practical manufacturing quality-control workflows rather than only returning an AI prediction.

- PASS – inspection is considered acceptable.
- FAIL – inspection indicates a defect requiring rejection/action.
- REVIEW – result requires quality-engineer verification.

6. Inspection History

A dedicated Inspection History page was implemented to provide a searchable view of previous inspection records. The history includes identifiers, filenames, product categories, defect types, quality results, severity and date/time.

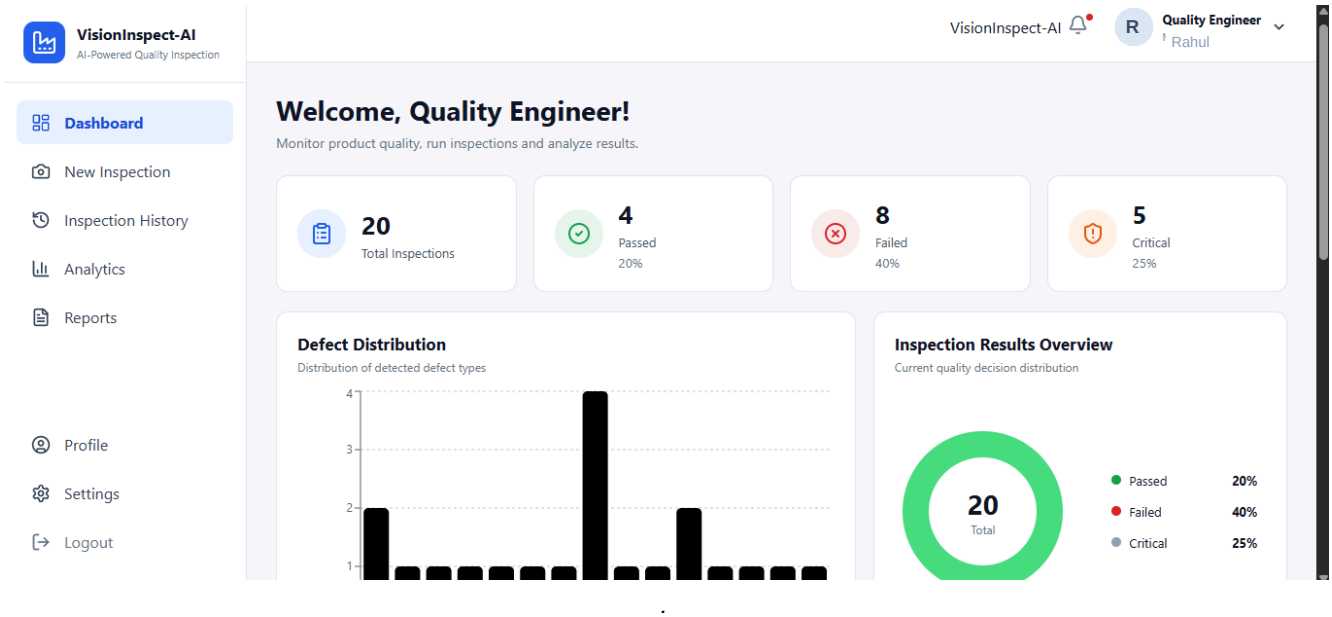


7. Manufacturing Analytics Dashboard

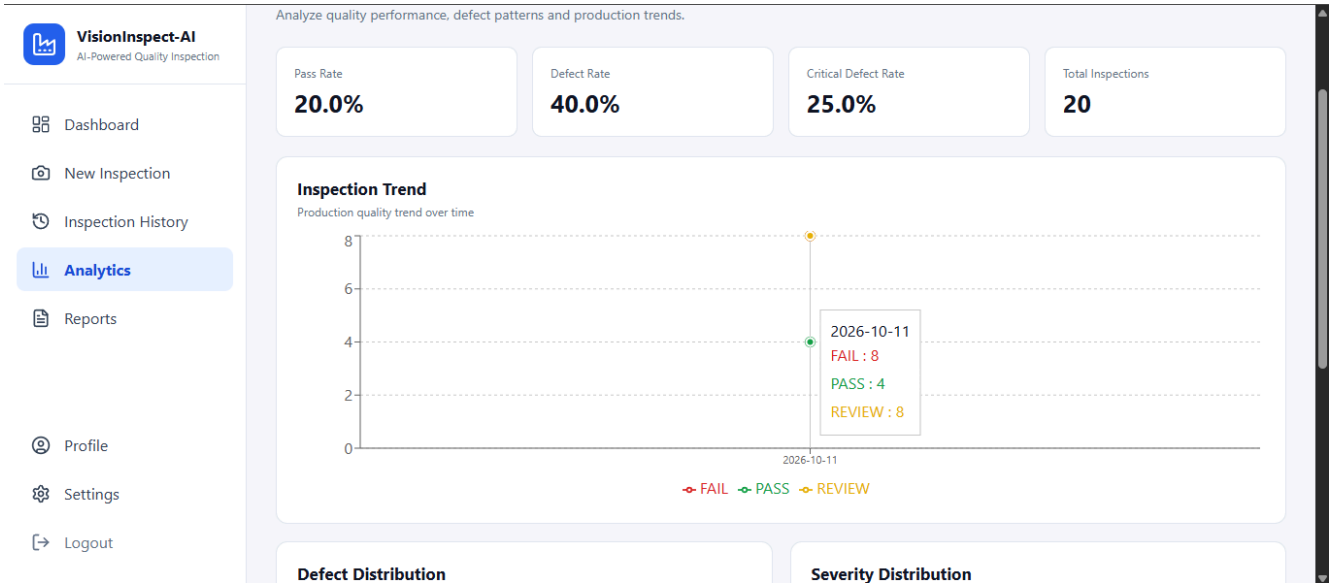
The analytics module provides an overview of production-quality performance. The dashboard summarizes total inspections, pass rate, defect rate and critical defect rate, and provides visualizations of inspection trends and defect distributions.

- Pass Rate
- Defect Rate
- Critical Defect Rate
- Total Inspections
- Inspection Trend
- Defect Distribution
- Severity Distribution

Quality Engineer Dashboard



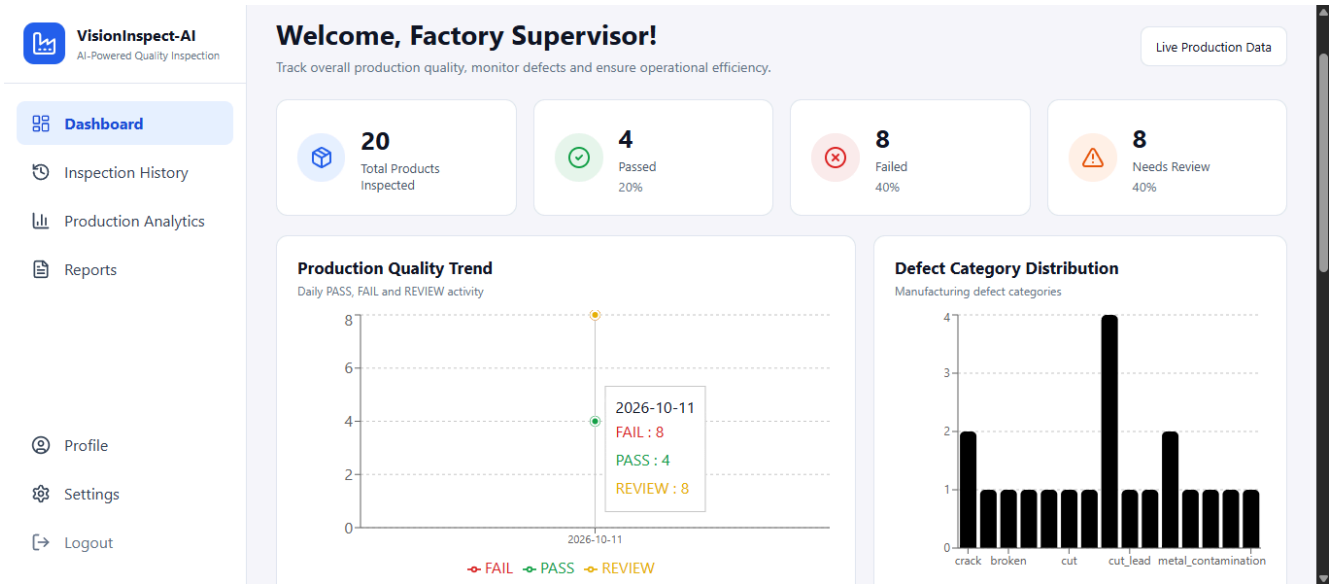
Analytics Dashboard:



8. Factory Supervisor View

A Factory Supervisor-oriented dashboard was included to present production quality information at an operational level. It summarizes products inspected, passed, failed and items requiring review, along with production trends and defect-category distribution.

Factory Supervisor Dashboard:

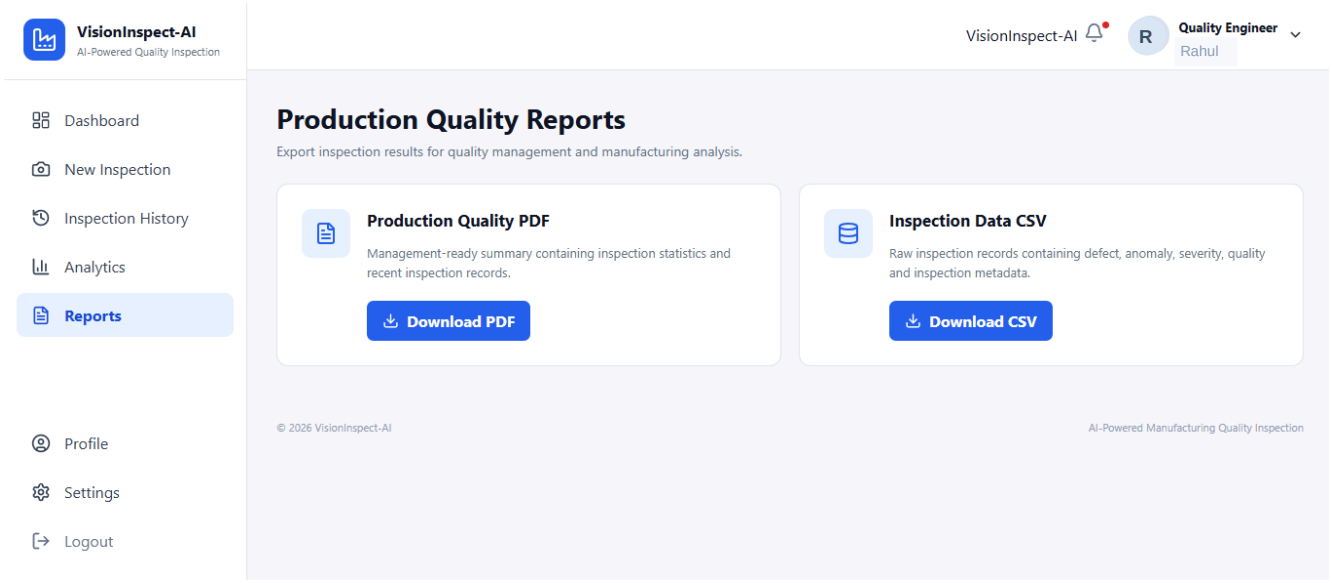


9. Reporting

The Reports module provides downloadable production-quality information. The application supports generation of a production quality PDF summary and inspection-data CSV output.

- Production Quality PDF report.
- Inspection Data CSV export.
- Reports contain inspection statistics and inspection metadata.

Reports Page:



10. Backend Analytics APIs

- GET /api/analytics/statistics
- GET /api/analytics/trend
- GET /api/analytics/defect-distribution

- GET /api/analytics/severity-distribution
- GET /api/analytics/history

11. Milestone 3 Outcome

Milestone 3 transformed VisionInspect AI from a basic AI inspection application into a broader quality-control platform. Users could view classification and severity information, make quality decisions, review historical inspections, analyze production trends and export reports.

12. Conclusion

Milestone 3 successfully implemented the major quality-management features required for the project. The system was ready for final validation, containerization, cloud deployment and end-to-end demonstration in Milestone 4.